

VRAJ LATHIYA

Leicester, UK

PROFESSIONAL SUMMARY

Adaptive Computer Science professional with a focus on AI Engineering and scalable software solutions. Proficient in integrating LLMs and RAG architectures into production environments while maintaining core competencies in full-stack development. Committed to building efficient, ethically-aligned systems that bridge complex machine learning research with user-centric applications.

PROJECTS

Fast-API RAG Chatbot | Personal Project

April 2026 – May 2026

- Built a high-performance RAG chatbot using Next.js, Vercel AI SDK, LangChain, and Groq with low-latency streaming and clickable source citations.
- Engineered an automated data ingestion pipeline to scrape, chunk (1,000 tokens / 200 overlap), and embed 130+ pages of API documentation into a Chroma vector database.
- Implemented multi-turn conversational memory using Llama-3.1-8B to dynamically rewrite context-aware follow-up queries.
- Achieved high-accuracy generation via Groq's Llama-3.1-70B model, significantly reducing response latency.

AI-Powered Poetry Generator | University of Leicester

June 2025 – August 2025

- Architected a full-stack social platform using Django and DRF with a normalised database schema spanning 7 key entities.
- Engineered an iterative AI refinement loop using GPT-4 and GPT-5 APIs to increase user agency through co-creation.
- Implemented a multi-mode discovery system with 3 feed types and AJAX-powered filtering.
- Developed a dual-translation engine offering 'Poetic' and 'Accurate' modes across 6+ languages.
- Enhanced security with JWT authentication including token rotation and UUID-based email verification.

EXPERIENCE

Student Support Specialist | Cosmic People

January 2025 – May 2026

Leicester, UK

- Managed and delivered comprehensive academic support for students with sensory impairments and complex learning needs across undergraduate and postgraduate disciplines.
- Acted as a Sighted Guide and scribe, enabling students with visual impairments to navigate campus and participate fully in seminars.
- Served as a formal Invigilator, Scribe, and Reader for university-level examinations.
- Managed student workflows via Blackboard LMS, tracking assessment deadlines and coordinating digital submissions.
- Tailored note-taking styles to individual needs, including formatting for screen readers and phonetic cues.

Junior Technician | Rushabh Infoworld

August 2020 – August 2021

Rajkot, Gujarat, India

- Provided technical support by identifying and addressing customer needs with timely, effective solutions.
- Assembled custom-built computers based on individual client requirements, ensuring optimal hardware configurations.

- Diagnosed a wide range of hardware and software issues, including malfunctioning components and system crashes.
- Communicated complex technical concepts to non-technical customers, improving satisfaction and retention.

EDUCATION

MSc in Advanced Computer Science | University of Leicester 2024 – 2025
Dissertation: AI-based Poetry Generator with community features.

Bachelor of Computer Applications | Atmiya University 2021 – 2024
Grade: 8.6 GPA · Specialisation: Java and Python · Relevant Coursework: Cyber Security and Networking

CERTIFICATIONS

- CPD Sighted Guide, Disability Awareness & Safeguarding 2025
- ASP.NET MVC for Beginners — Great Learning Academy 2024
- Cyber Security — Atmiya University 2023
- Fundamentals of Networking (CISCO) — Atmiya University 2022

INVOLVEMENT

Participant — Google GDSC Hackathon (Atmiya University) March 2024
Collaborated with a team to develop a functional prototype using Java and SQL within the hackathon timeframe.

Social Worker — Sadbhavna Vridhhashram June 2022
Led a city-wide initiative to distribute 250+ eco-friendly bird nests and promoted environmental awareness.

SKILLS

AI / Data Science: RAG Architecture · LLM Integration · NLP · Prompt Engineering
Languages: Python · JavaScript · SQL
Frameworks & Tools: Django · Next.js · FastAPI · LangChain · Ollama · ChromaDB · Git
Soft Skills: Punctuality · Critical Thinking · Multilingual (English, Hindi, Gujarati)